Chinese ADR Long-Short: Review and Recommendations

Review of the adr_longshort repository as of 18 October 2026
19 October 2026

Summary

We recommend **not trading the strategy as specified**. Keep it as research, and paper-trade it from September 2026.

The forecast is real. The ranking beats every random sort we tried, and it is good at finding Chinese small caps that are about to fall.

The profit is not. It sits in shorts that retail lenders price at roughly what they earn. The backtest's universe also used information from the future.

After correction: we fixed the data and priced borrow from observed fees. The $500,000 account then ends at **$0.52–0.69m** over 2010–26, against $0.61m in Treasury bills.

Four changes would make the evidence honest and show whether anything is left to trade:

1. **Fix the universe.** Measure liquidity on typical days, apply the $1 test to the price as traded, drop the two filters that use future data, and remove reused ticker histories.
2. **Price borrow and spreads per name.** The schedule charges 6–7% a year. Observed IBKR fees on the model's shorts run at a median of 58–67%.
3. **Count the trials.** At the repository's own 70 trials, the corrected deflated Sharpe is 0.05. At the 322 we can count, it is 0.005.
4. **Treat the short leg as a toxicity premium.** Risk-manage it as one. The tail is real and comes from squeezes, not data errors.

Section 1 covers what holds up. Section 2 gives the four changes. Section 3 adds smaller suggestions. Section 4 lists our questions, and section 5 the concerns to keep in view. The appendix holds the evidence.

Conventions.

- Figures run January 2010 to August 2026.
- **Account** means the repository's $500,000 simulation: 10% volatility target, 20 bp a side, tiered borrow, cash interest.
- **Unit book** means $1 long and $1 short, monthly.
- Sharpe ratios are daily unless marked monthly.
- **Shipped** is the repository as cloned. **Corrected** is the data of section 2.1.

1. What holds up

- **The pipeline reproduces.** The tests pass, and a full refit matches the shipped forecasts to 1e-16.
- **The ranking has information.**
  - None of 500 random rankings, run through the same rules and costs, matches the corrected account.

  - Rank IC is 0.089 (t 7.6) on corrected data.
  - The worst-ranked fifth of names trading above $3m a day falls 18.7% a year.
- **The forest earns its place where it trades.**
  - Huber regression has the higher average IC (0.12 against 0.09). But its account ends at $0.74m against the forest's $1.61m.
  - The forest is better at the extremes, which are the names it actually trades. Forest minus Huber has paired t 2.1.
- **The two-year freeze is not a weakness.** Every way of training on more recent data does worse. Retraining through last year loses at t −2.3; the recent years are the noisiest.
- **The cleaning does not flatter the P&L.**
  - 777 of the 871 removals are verified data errors: price-scale flips, unadjusted splits, and moves a second source contradicts.
  - Putting back the real and unverifiable moves changes the account by less than $10k, with no margin call.
- **The book is market-neutral.** Beta to PGJ is about zero in every corrected variant.
- **It is independent of the trend book.** The correlation with trend-following funds is 0.06–0.08.
- **The long leg is a good hedge over the full period.** It earns about 11% a year of alpha over the volume-weighted universe, and beats every ETF hedge we tried (account 0.68 against 0.42 for PGJ).

2. Changes we recommend

2.1 Fix the universe before anything else

Four rules let the backtest see the future or trade names it couldn't have traded:

- **The $3m short floor uses mean daily volume.** One pump day can lift a name over it. 13% of the shipped short book's name-months, and 21% since 2021, cleared $3m only on the mean, trading under $1.5m on a typical day. GMM qualified on a single $70m day. These names fell 10.6% a month against 2.5% for the rest.
- **The $1 test and a "$5,000" exclusion use split-adjusted prices.** The exclusion dropped 55 serial reverse-splitters for their whole history, using their future price path. All 55 lost 99–100%.
- **25 names carry another company's history under the same ticker.** NEW, CIH, WEI and HMI go back to 2000. The backtest's worst day (18 July 2011, −7.8%) was "NEW", a different company, rising 150% in the short book.
- **The scored pool requires ten trading days of next-month data.** In practice this mostly guards against the price archive ending in April 2020, so it matters little.

Fixing all four lowers the result, mostly through the volume rule:

	Shipped	Corrected
Final equity	$1.81m	$1.34m
Sharpe, daily / monthly	0.75 / 1.01	0.62 / 0.69
Worst drawdown / worst month	−15.0% / −4.6%	−14.7% / −9.6%

	Shipped	Corrected
Unit Sharpe 2010–15 / 2016–20 / 2021–26	1.40 / 1.18 / 1.50	1.13 / 0.88 / 0.96
Bootstrap 90% interval, monthly Sharpe	0.62–1.40	0.36–1.03

Each fix measured on its own: Appendix A.

We suggest running every later test on the corrected universe. Our own tests did.

2.2 Price borrow and spreads per name

What the backtest assumes. Borrow is charged by volume tier: 1% to 35% a year, averaging 6–7%.

What we observed. iBorrowDesk publishes IBKR's fee and shares available for the last 12 months. Over October 2025 to October 2026, the names this model shorts cost:

	Observed IBKR fee
Median / mean	67% / 157% a year
90th percentile	460% a year
Share above the "harsh" schedule	63%
Share above 100% a year	43%
Held longs, for contrast	median 0.4%

150 short name-months with an observed fee. Volume tiers don't predict the fee; listing age and the size of the collapse do.

The lender takes the profit. In the six months when every short had an observed fee (March–August 2026), the short leg made 17.9% a month gross. Observed fees turned that into −1.5% a month.

The fee model. It is fitted to the observed fees (cross-validated R^2 0.59) and puts the average for the shorts at about 37% a year over 2010–26.

Unavailable names carry the profit. Names that fail an availability rule we stated in advance (too few shares, fee above 100%, or on a threshold list) carry 40–50% of the short leg's profit, and 88% in 2025–26.

Trading costs. A one-cent tick on today's shorts (median price $1.95) is already 45–100 bp, and high-low spread estimates put the short leg at 40–110 bp a side. The backtest charges 20 bp.

Together:

Corrected account	Final	CAGR	Sharpe	Sharpe 2021–26
As built (20 bp, tiered borrow)	$1.34m	6.1%	0.62	0.79
20 bp, observed or imputed IBKR fees	$0.91m	3.7%	0.40	0.36
Half the spread estimate, IBKR fees	$0.68m	1.8%	0.23	0.13
…and skip unavailable names	$0.69m	2.0%	0.24	0.03
…and rank shorts net of their fee	$0.65m	1.6%	0.21	−0.11
Full spread estimate, IBKR fees, skip unavailable	$0.52m	0.2%	0.07	−0.15
Treasury bills less 0.5%	$0.61m	1.2%		

Per-name costs: Abdi-Ranaldo half-spread, floored at half a tick on the traded price, plus $0.0035 a share and square-root impact. Appendix G.

Ranking shorts net of their fee doesn't rescue it. It cuts the fee bill from 19% to 4% a year, but the alpha leaves with the expensive names.

We suggest **pricing borrow from a broker's history before any capital goes in**. The question to settle is narrow: at what fee could the model's shorts actually have been borrowed in 2021–24?

2.3 Count the trials, and fix the specification in advance

What the documents claim. The deflated Sharpe is 0.83 at 13 trials and 0.33 at the repository's own 70.

The count behind those figures is low. It covers models, not the choices of features, universe and floor. Our own review logged 252 more variants; most were diagnostics of choices the author had already made.

Account (monthly Sharpe)	13 trials	70 trials	322 trials
Shipped (1.01)	0.83	0.33	0.07
Corrected (0.69)	0.36	0.05	0.005

Probability that the true Sharpe is above zero after allowing for the number of trials (Bailey and López de Prado), using the repository's dispersion of trial Sharpes.

Our own candidate confirms the problem. We wrote one down before running it: the corrected data plus dropping five features that looked redundant. It scored 0.50 against 0.62 for the unchanged model. The feature gains we had seen were noise.

Choosing the model inside the protocol doesn't help either. Picking forest, Huber or OLS each December by validation was the worst option tested, because the choice flips almost every year.

We suggest:

- Fix the specification now.
- Count every look in the documents.
- Use the months from September 2026 as the only clean test.

2.4 Treat the short leg as a toxicity premium

The documents explain the edge as capacity: borrow and liquidity keep large funds out. The evidence points more to compensation for bearing short-side risk.

- **The alpha lives in the names that are hard to borrow.** Ranking shorts net of their fee, or weighting them by inverse volatility, cuts the alpha more than it cuts costs or tails (paired t −1.3 to −2.8).
- **The tail is real.**
  - In November 2020, the Chinese EV mania, the short book rose 55% in a month (CBAT +228%, KXIN +104%) and the account lost 12.2%.
  - STAK rose 809% intraday in July 2026, confirmed by Nasdaq's data.
  - A +200% day on the largest short costs 7% of equity in a typical month, up to 13%, and 16% at the five-name minimum. That is about three times the worst month the shipped backtest shows.
- **Hong Kong behaves the way short-sale constraints predict.**

  - On HKEX, the worst-ranked fifth underperforms by 22–26% a year among names that can't be shorted, and by roughly nothing among names that can.
  - Run outside China, the same pipeline earns mostly short-market exposure, and the profitable version exists only in China.

Reg T margin isn't the risk. It never binds, because three-quarters of equity sits in cash. The risks are recalls, buy-ins and a single squeeze.

We suggest choosing the risk budget on grounds outside this backtest. Two options:

- accept the tail and size small;
- or cap single names and give up some alpha.

Higher fixed exposure only raises the drawdown: fixed 50% a side reaches −32%.

3. Smaller suggestions

3.1 Keep the freeze, and say why

The model trading in 2026 never saw data after 2023. That looks like a crack, but it isn't one. Every way of training on more recent data scored worse:

- one-year validation: account 0.38;
- retraining through last year: 0.34;
- quarterly refits: 0.48;
- against 0.68 for the freeze.

Only recency weighting tied it (0.72, t +0.1). The documents should state this test instead of leaving the freeze unexplained.

3.2 Clean the P&L only for verified errors

The +300% rule deletes the up-day of a squeeze that doesn't reverse, but keeps the fall that follows. AMTD Digital's September 2022 shows as −90.5% instead of the real −58%.

No corrected-book position was affected. Still, training may use a filter chosen on validation, while the P&L should remove only verified errors. Our audit script logs and classifies every removal.

3.3 Describe the features by what they do in the traded set

- **zero_1m ranks first by the forest's own importance but carries no signal in the traded set.** Its rank IC among names above $3m is −0.007 (t −0.5), and it is exactly zero in 81% of those rows.
- **illiq_1m, maxret_1m and minret_1m are redundant.** The other features explain 80–92% of each, and removing them is neutral.
- **A four-feature trend forest** (high52 and three momentum terms) matches all 16 on the corrected data.

We would not change the model on this evidence; every gain is inside the seed noise. We suggest reporting importance by permutation on the liquid names, not by impurity.

3.4 Present the history in its regimes

- **2010–15 is the least reliable period.** 266 China- and HK-based issuers delisted from US exchanges in 2009–26, and only 17 have prices in the data. That includes the whole 2011–12 reverse-merger fraud wave.
- **The long leg's alpha is concentrated in a few years.** Long minus PGJ was +169% in 2020, +70% in 2017 and +50% in 2013, and about nil since 2021.
- **Report the daily Sharpe as the headline.** The shipped monthly Sharpe (1.01) overstates the daily one (0.75), because daily returns reverse within the month.

3.5 Correct the data notes

- **The Kaggle names are mislabelled.** `data/README.md` describes them as "delisted before April 2020". They are names listed in April 2020 whose later months are missing.
- **Hong Kong-based listings are excluded.** AMTD Digital, which the model would have shorted in 11 months, is one example. The exclusion should be stated or the names added.

3.6 Present sizing as a volume knob

The volatility target changes return, not quality:

- the 5%, 10%, 15% and 20% targets give Sharpe 0.71, 0.62, 0.59 and 0.58;
- fixed 15–50% a side gives 0.72–0.61.

The choice should rest on the drawdown the account can bear: −7.6% at a 5% target, −14.7% at 10%, −28.5% at 20%.

4. Questions for the author

1. Was the $3m floor meant to use mean daily volume? Did you check how many shorts qualify on one or two days of trading?
2. Was it known that the "$5,000 adjusted price" rule uses each name's full future price path?
3. How were the 13 and 70 trials counted? Do they include the choices of features, universe and short floor?
4. Do you have broker records (fees, locates, recalls) for names shorted since 2021?
5. Why are Hong Kong-based companies such as AMTD Digital outside the universe?
6. What should the strategy do in a month like November 2020, when the whole short book squeezes at once?
7. Were the 2011–12 reverse-merger names ever in a research universe, and how did they behave?

5. Concerns to keep in view

- **Survivorship cannot be signed with free data.** Fraud collapses would have been gains for the short leg on paper, but were often halted for months with borrow accruing. Take-privates, 123 of them, paid 20–40% premiums on the beaten-down names the model shorts.
- **Borrow before 2025 is modelled, not observed.** IBKR's retail fees are also a lower bound: most other brokers charge more or can't locate these names.

- **Spreads are estimates.** High-low estimators confuse volatility with spread. The true cost lies inside the range we give, not at one end of it.
- **Capacity is small even before borrow.** A typical short reaches 1% of daily volume at about $5.4m of equity; the 90th-percentile short does at $1.3m.
- **The supply of the names the short leg needs may be shrinking.** The summary notes Nasdaq's new minimum IPO size for China-based companies and the SEC's cross-border task force. We did not verify those.

Appendix

A. The universe, one fix at a time

	Shipped	Real-price $1 test, no $5k rule	No next-month filter	Median volume	All four	+ ticker reuse
Unit long-short Sharpe	1.34	1.24	1.32	1.01	1.11	0.98
Short leg, %/yr	−28.5	−29.7	−28.8	−10.4	−20.1	−19.0
Account Sharpe	0.75	0.69	0.75	0.56	0.68	0.62
Account 2021–26	0.98	0.83	0.98	0.50	0.68	0.80
Worst month	−4.6%	−8.5%	−4.3%	−7.4%	−12.2%	−9.6%
Final equity	$1.79m	$1.67m	$1.80m	$1.23m	$1.61m	$1.34m

yfinance split events: 360 (333 reverse); 9 never absorbed into Yahoo's adjusted prices; GMM's November 2024 split missing from Yahoo's list.

B. The cleaning audit

Verdict	Removals	Main reasons
Verified error	777	mixed price scales 687, unlisted splits 45, second source disagrees 36, listed splits 9
Real move	46	Nasdaq confirms 38, volume confirms 8
Unknown	48	archive names without a second source

Book	Shipped cleaning	Only verified errors removed	No cleaning
Shipped	$1.806m, 0.75	$1.798m, 0.75	$1.742m, 0.72
Corrected	$1.608m, 0.68	identical	identical

12 removals fell inside shipped positions: CIH scale flips (longs), AIXI (Yahoo error, short), STAK +602% (real, short). None fell inside corrected positions.

C. Forest, Huber, OLS, blends (corrected)

Model	IC	Unit Sharpe	Account	Final	Top − bottom quintile, ≥ $1m / ≥ $3m	Bottom quintile, ≥ $3m
Random forest	0.09	1.13	0.68	$1.61m	34.3 / 40.6	−18.7%
Huber	0.12	0.60	0.27	$0.74m	24.0 / 21.9	−9.3%
OLS	0.09	0.73	0.38	$0.88m	27.0 / 28.0	−6.6%
Blend RF + OLS	0.10	1.02	0.63	$1.38m	34.1 / 36.0	−13.9%
Chosen each December	0.10	0.77	0.40	$0.95m		

Paired t, monthly net: forest − Huber 2.12, forest − OLS 2.19 (about 2.0 ± 0.45 across seeds); forest − blends 0.45–0.78.

D. Refit timing (corrected)

Training scheme	Unit Sharpe	Account	Final	t vs freeze
Freeze: train to Y−3 (shipped)	1.13	0.68	$1.61m	—
One-year validation: train to Y−2	0.85	0.38	$0.88m	−2.04
Retrain through Y−1	0.77	0.34	$0.85m	−2.33
Rolling 8 years	1.12	0.61	$1.37m	−0.51
Recency-weighted	1.23	0.72	$1.70m	+0.10

Training scheme	Unit Sharpe	Account	Final	t vs freeze
Quarterly refit	0.93	0.48	$1.07m	−1.36
Chosen each December	1.16	0.65	$1.47m	−0.34

E. Features (corrected)

Variant	Unit Sharpe	Account	Worst month	Note
All 16	1.12 ± .12	0.68 ± .08	−12.2%	baseline, seeds 0–4
high52 + three momentum	1.33	0.76	−8.0%	fixed hyperparameters
Without zero_1m	1.12 ± .04	0.68 ± .03	−6.4%	2021–26 account 0.48 vs 0.68
Without the redundant five	1.21 ± .08	0.73 ± .07		paired t −0.06
Pre-registered 11-feature candidate	0.90 (monthly)	0.50	−10.2%	final data, Appendix J

Share of each feature explained by the other 15 (R^2): illiq_1m 0.92, maxret_1m 0.84, minret_1m 0.80, beta_12m 0.39, age 0.20, zero_1m 0.17.

F. Construction and the hedge (corrected)

Book	Unit net (base / harsh)	Account	Final	Drawdown
Short only	0.26 / 0.10	0.21	$0.66m	−31%
Short + long leg (baseline)	0.81 / 0.61	0.68	$1.61m	−17%
Short + volume-weighted universe	0.72 / 0.50	0.66	$1.49m	−19%
Short + PGJ (1× / beta-matched)	0.51 / 0.50	0.42 / 0.40	$1.01m	−31%
Short + FXI	0.39	0.30	$0.80m	−31%
Inverse-volatility shorts	0.65 (t −2.80)	0.58	$1.30m	−20%
Borrow-aware ranking (base)	0.65 (t −1.33)	0.58	$1.32m	−16%

2010–20 → 2021–26 (corrected)	Value
Shorts listed under 24 months	48% → 22%
Short beta to PGJ	1.12 → 1.04
Long-leg correlation with PGJ	0.80 → 0.79
Long minus PGJ, %/yr	+32.2 → +5.5

The micro-IPO explanation for the hedge breaking is mostly rejected: betas and correlations are flat; the long leg's stock-picking alpha decayed.

G. Borrow and costs

Unit short leg, %/yr (Sharpe)	Shipped	Corrected
Gross	31.7 (0.74)	22.9 (0.50)
Tiered schedule	25.0 (0.59)	15.9 (0.35)
Observed or imputed IBKR fee	2.3 (0.06)	3.1 (0.07)
Available names only	5.3 (0.12)	4.7 (0.10)
Share of short profit in unavailable names	50%	40%

Fee model: log IBKR fee on volume, listing age, 12-month return, price, idiosyncratic volatility, threshold-list share and fails-to-deliver; R^2 0.61 (cross-validated 0.59); capped at 511%.

H. Sizing, costs and borrow (corrected account)

Change	Sharpe	Change	Sharpe
5% / 10% / 20% target	0.71 / 0.62 / 0.58	Cost 10 / 20 / 40 / 60 bp	0.68 / 0.62 / 0.49 / 0.36
Fixed 15% / 50% a side	0.72 / 0.61	Borrow zero / base / harsh	0.79 / 0.62 / 0.43
250-day volatility	0.65	Flat borrow 15% / 25% / 40%	0.42 / 0.17 / −0.21

I. Other markets (identical frozen pipeline, monthly Sharpe)

Universe	Rank IC (t)	Long-short gross / net	Short leg alone	Beta	Bottom quintile − EW, %/yr
China (corrected)	0.089 (7.6)	1.22 / 0.87	0.51	−0.08	−18
US microcaps	0.055 (6.5)	0.64 / 0.20	0.39	−0.51	−13
HKEX small caps	0.050 (5.7)	0.54 / −0.07	0.00	−0.39	−7
Non-China Asian micro-listings (≈3 yrs)	0.128 (4.7)	too few names			−41 to −53

HKEX: bottom quintile −22% to −26% a year among names that can't be shorted, about 0 among shortable names; the flag is today's list, not point-in-time.

J. Final checks

	Shipped	Corrected, 16 features	Candidate, 11 features
Final equity	$1.81m	$1.34m	$1.08m
Sharpe, daily / monthly	0.75 / 1.01	0.62 / 0.69	0.50 / 0.56
Bootstrap 90%, monthly Sharpe	0.62–1.40	0.36–1.03	0.20–0.95
Random sorts reaching it (of 500)	0	0	0
Deflated Sharpe, 70 / 322 trials	0.33 / 0.07	0.05 / 0.005	0.02 / 0.001
Five-year fresh account	$852k	$776k	$720k

Full working notes and reproduction commands: reports/INSPECTION.md. Scripts and tables: research/ (local, uncommitted).